

Jorge Peña

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July 24, 2025

Education

Ph.D. in Applied Mathematics for the Human and Social Sciences

Faculty of Social and Political Sciences, University of Lausanne, Lausanne, Switzerland

2012

Master of Advanced Studies (M.A.S.) in Embedded Systems Design

University of Lugano, Lugano, Switzerland

2006

Electronics Engineering

Universidad Pontificia Bolivariana, Medellín, Colombia

2004

5-year degree

Employment History

Professor

Toulouse School of Economics
Department of Social and Behavioral Sciences

Toulouse, France
since September 2024

Assistant Professor

Toulouse School of Economics
Department of Social and Behavioral Sciences

Toulouse, France
September 2023 – August 2024

Guest Researcher

Max Planck Institute for Evolutionary Anthropology
Department of Human Behavior, Ecology and Culture

Leipzig, Germany
March 2023 – July 2023

Research Fellow

Institute for Advanced Study (IAS), University of Amsterdam

Amsterdam, Netherlands
September 2022 – February 2023

Assistant Professor

Institute for Advanced Study in Toulouse, University of Toulouse 1 Capitole

Toulouse, France
September 2017 – August 2022

Postdoctoral Researcher

Geomar Helmholtz Centre for Ocean Research Kiel

Kiel, Germany
September 2016 – August 2017

Postdoctoral Researcher

Max Planck Institute for Evolutionary Biology

Plön, Germany
September 2013 – August 2016

Postdoctoral Researcher

Faculty of Business and Economics, University of Basel

Basel, Switzerland
April 2012 – August 2013

Graduate Assistant

Faculty of Social and Political Sciences, University of Lausanne

Lausanne, Switzerland
September 2006 – August 2011

Fixed-term Part-time Lecturer

Faculty of Electronics Engineering, Universidad Pontificia Bolivariana

Medellín, Colombia
January 2004 – August 2005

Research

Working Papers

(24) Heifetz, A., Peña, J. Nash equilibria are extremely unstable in most games under the utility-taking gradient dynamics. <https://ideas.repec.org/p/tse/wpaper/130433.html>

Articles in Refereed Journals

* Equal contribution

- (23) Molho, C., **Peña, J.**, Singh, M., Derex, M. (2024). [Do institutions evolve like material technologies?](#) *Current Opinion in Psychology*, 60, 101913
- (22) Mullon, C., **Peña, J.**, Lehmann, L. (2024). [The evolution of environmentally mediated social interactions and posthumous spite under isolation by distance.](#) *PLOS Computational Biology*, 20(5): e1012071
- (21) **Peña, J.**, Heifetz, A., Nöldeke, G. (2024). [The shirker's dilemma and the prospect of cooperation in large groups.](#) *Theoretical Population Biology*, 155, 10-23
- (20) **Peña, J.**, Nöldeke, G. (2023). [Cooperative dilemmas with binary actions and multiple players.](#) *Dynamic Games and Applications* 13, 1156-1193
- (19) Cooper, G.A., Liu, M., **Peña, J.**, West, S. A. (2022). [The evolution of mechanisms to produce phenotypic heterogeneity in microorganisms.](#) *Nature Communications* 13, 195
- (18) González-Forero, M.*, **Peña, J.*** (2021). [Eusociality through conflict dissolution.](#) *Proceedings of the Royal Society B*. 288: 2021038620210386
- (17) Nöldeke, G., **Peña, J.** (2020). [Group size and collective action in a binary contribution game.](#) *Journal of Mathematical Economics* 88, 42-51
- (16) **Peña, J.**, Nöldeke, G., Puebla, O. (2020). [The evolution of egg trading in simultaneous hermaphrodites.](#) *The American Naturalist* 195(3), 524-533
- (15) Estrela, S., Libby, E., Van Cleve, J., Débarre, F., Deforet, M., Harcombe, W. R., **Peña, J.**, Brown, S. P., Hochberg, M. E. (2019). [Environmentally mediated social dilemmas.](#) *Trends in Ecology & Evolution*, 34(1), 6-18
- (14) **Peña, J.**, Nöldeke, G. (2018). [Group size effects in social evolution.](#) *Journal of Theoretical Biology*, 457, 211-220
- (13) Pichugin, Y., **Peña, J.**, Rainey, P., Traulsen, A. (2017). [Fragmentation modes and the evolution of life cycles.](#) *PLOS Computational Biology* 13(11): e1005860
- (12) dos Santos, M., **Peña, J.** (2017). [Antisocial rewarding in structured populations.](#) *Scientific Reports* 7: 6212
- (11) **Peña, J.**, Wu, B., Arranz, J., Traulsen, A. (2016). [Evolutionary games of multiplayer cooperation on graphs.](#) *PLOS Computational Biology* 12(8): e1005059
- (10) Nöldeke, G., **Peña, J.** (2016). [The symmetric equilibria of symmetric voter participation games with complete information.](#) *Games and Economic Behavior*, 99, 71-81
- (9) **Peña, J.**, Wu, B., Traulsen, A. (2016). [Ordering structured populations in multiplayer cooperation games.](#) *Journal of the Royal Society Interface*, 13: 20150881
- (8) **Peña, J.**, Nöldeke, G. (2016). [Variability in group size and the evolution of collective action.](#) *Journal of Theoretical Biology*, 389, 72-82
- (7) **Peña, J.**, Nöldeke, G., Lehmann, L. (2015). [Evolutionary dynamics of collective action in spatially structured populations.](#) *Journal of Theoretical Biology*, 382, 122-136
- (6) **Peña, J.**, Lehmann, L., Nöldeke, G. (2014). [Gains from switching and evolutionary stability in multi-player matrix games.](#) *Journal of Theoretical Biology*, 346, 23-33
- (5) **Peña, J.**, Rochat, Y. (2012). [Bipartite graphs as models of population structures in evolutionary multiplayer games.](#) *PLOS ONE*, 7(9): e44514
- (4) **Peña, J.** (2012). [Group-size diversity in public goods games.](#) *Evolution*, 66, 623-636
- (3) Buesser, P., **Peña, J.**, Pestelacci, E., Tomassini, M. (2011). [The influence of tie strength on evolutionary games on networks: an empirical investigation.](#) *Physica A*, 390, 4502-4513
- (2) **Peña, J.**, Pestelacci, E., Berchtold, A., Tomassini, M. (2011). [Participation costs can suppress the evolution](#)

of upstream reciprocity. *Journal of Theoretical Biology*, 273, 197-206

- (1) Peña, J., Volken, H., Pestelacci, E., Tomassini, M. (2009). Conformity hinders the evolution of cooperation on scale-free networks. *Physical Review E*, 80, 016110

Articles in Refereed Proceedings in Computer Science

- (9) Peña, J., Pestelacci, E., Tomassini, M., Volken, H. (2009). Conformity and network effects in the prisoner's dilemma. In *Proceedings of the IEEE Congress on Evolutionary Computation, 2009*, pp. 506-513
- (8) Montes De Oca, M. A., Peña, J., Stützle, T., Pinciroli, C., Dorigo, M. (2009). Heterogeneous particle swarm optimizers. In *Proceedings of the IEEE Congress on Evolutionary Computation, 2009*, pp. 698-705
- (7) Peña, J. (2008). Conformist transmission and the evolution of cooperation. In *Artificial Life XI: Proceedings of the Eleventh International on the Simulation and Synthesis of Living Systems*, pp. 458-465. MIT Press, Cambridge, MA
- (6) Peña, J. C., Peña, J., Upegui, A. (2008). Evolutionary graph models with dynamic topologies on the ubichip. In *Evolvable Systems: From Biology to Hardware. Proceedings of the 8th International Conference, ICES 2008. Lecture Notes in Computer Science* 5216, pp. 59-70
- (5) Peña, J. (2008). Simple dynamic particle swarms without velocity. In *Ant Colony Optimization and Swarm Intelligence. Proceedings of the 6th International Conference, ANTS 2008. Lecture Notes in Computer Science* 5217, pp. 144-154
- (4) Peña, J. (2008). Theoretical and empirical study of particle swarms with additive stochasticity and different recombination operators. In *Proceedings of the 10th Annual Conference on Genetic and Evolutionary Computation, GECCO '08*, pp. 95-102. ACM, New York, NY
- (3) Peña, J., Upegui, A. (2007). A population-oriented architecture for particle swarms. In *Proceedings of the Second NASA/ESA Conference on Adaptive Hardware and Systems, 2007*, pp. 563-570
- (2) Peña, J., Upegui, A., Sanchez, E. (2006). Particle swarm optimization with discrete recombination: an online optimizer for evolvable hardware. In *Proceedings of the First NASA/ESA Conference on Adaptive Hardware and Systems, 2006*, pp. 163-170
- (1) Peña, J., Vanegas, M., Valencia, A. (2006). Digital hardware architectures for Kohonen's self-organizing feature maps with exponential neighboring function. In *Proceedings of the IEEE International Conference on Reconfigurable Computing and FPGAs, 2006*, pp. 1-8

Fellowships, grants and awards

Institute for Advanced Study (IAS) Research Fellowship. September 2022 - February 2023

IAST Multidisciplinary Prize (with M. Derex, C. Molho, and M. Singh): "The Evolution of Institutions for Collective Action"

Swiss National Science Foundation – Early Mobility Postdoc Fellowship. PBLAP3-145860: "Multiplayer game theory and the evolution of collective action." 42,500 CHF. 2013–2014

Swiss Government Excellence Scholarship for Foreign Scholars and Artists. 21,840 CHF. 2005–2006

Beca Municipio de Medellín a los Mejores Bachilleres. 1999–2003

Presentations

Invited Seminars

Department of Biology, University of Pennsylvania, May 15, 2025

Theory Tea, Department of Ecology and Evolutionary Biology, Princeton University, May 13, 2025

Séminaire parisien de Théorie des Jeux, March 10, 2025

Department of Experimental and Applied Psychology, VU Amsterdam, May 16, 2024

Theoretical Biology and Bioinformatics, Utrecht University, May 14, 2024

Department of Human Behavior, Ecology and Culture, Max Planck Institute for Evolutionary Anthropology, May 9, 2023

VU Amsterdam, Expert Workshop, May 17, 2023

Leiden University, Conflict and Cooperation Laboratory, January 31, 2003

University of Münster, Münster Graduate School of Evolution, The Growth of Evolutionary Thought (public lecture series), January 23, 2023

University of Amsterdam, Institute for Advanced Study, Kick-off lecture IAS fellowship, November 10, 2022

Max Planck Institute for Evolutionary Biology, Dynamics of Social Behavior Research Group, July 14, 2022

University of Graz, COLIBRI seminar (online), April 1, 2022

Queen Mary University of London, Complex Systems Seminar (online), December 2020

Sorbonne Université, Laboratoire Jean Perrin, February 2019

University Claude Bernard Lyon, Laboratoire de Biométrie et Biologie Évolutive, October 2018

Aarhus University, Department of Bioscience, April 2018

Toulouse School of Economics, MAD-Stat seminar series, March 2018

University of St Andrews, School of Biology, July 2017

Monash University, Faculty of Information Technology, November 2016

Santa Fe Institute, June 2016

Institute for Advanced Study in Toulouse, May 2015

University of Lausanne, Department of Ecology and Evolution, June 2013

Max Planck Institute for Evolutionary Biology, Research Group for Evolutionary Theory, May 2013

Universidad Carlos III de Madrid, Department of Mathematics, May 2012

Université Pierre & Marie Curie, Séminaires Ecologie et Evolution, May 2012

Universidad Pontificia Bolivariana, Escuela de Ingenierías, February 2012

Max Planck Institute for Evolutionary Biology, Research Group for Evolutionary Theory, November 2011

Universidad Pontificia Bolivariana, Escuela de Ingenierías, March 2011

Universidad EAFIT, Departamento de Ciencias Matemáticas, March 2011

Universidad de Antioquia, Departamento de Matemáticas, March 2011

Workshops and Conferences

Mathematics and Social Sciences Workshop, 2025, talk

Developments in Evolutionary Theory Workshop, Max Planck Institute for Evolutionary Biology, 2024, talk

ICSD 2024 (Leiden), International Conference on Social Dilemmas, talk

FRESH 4th Annual Conference (Paris), talk

ICSD 2022 (Copenhagen), International Conference on Social Dilemmas, talk

SMB 2021, Virtual Society for Mathematical Biology Meeting, online talk

ESEB 2019 (Turku), Congress of the European Society for Evolutionary Biology, talk

MMEE 2019 (Lyon), Mathematical Models in Ecology and Evolution Conference, talk

Edinburgh Napier University, Workshop on Evolution and Dynamics of Institutions, May 2019, talk

EHBEA 2019 (Toulouse), 14th Annual Conference for the European Human Behaviour and Evolution Association, poster

Evolution 2018 (Montpellier), II Joint Congress on Evolutionary Biology, talk
 SHOW 2018 (Paris), Simultaneously Hermaphroditic Organisms Workshop, talk
 MMEE 2017 (London), Mathematical Models in Ecology and Evolution Conference, talk and poster
 Max Planck Institute for Evolutionary Biology (Plön), Workshop on Conflict, Competition, Cooperation & Complexity Workshop, January 2016, talk
 MMEE 2015 (Paris), Mathematical Models in Ecology and Evolution Conference, talk
 Institute for Advanced Study in Toulouse, Toulouse Economics and Biology Workshop, June 2015, poster
 ECMTB 2014 (Gothenburg), 9th European Conference on Mathematical and Theoretical Biology, talk
 Max Planck Institute for Evolutionary Biology (Plön), Spatial Human Cooperation Meeting, May 2014, talk
 Institute for Advanced Study in Toulouse, Toulouse Economics and Biology Workshop, May 2014, poster
 Max Planck Institute for Evolutionary Biology (Plön), 6th Workshop on Theoretical Biology, talk
 Inclusive Fitness and Game Theory Workshop (Arolla), May 2013, talk
 University of Basel, 4th Bernoulli Workshop in Economics and Psychology, June 2012, talk
 MMEE 2011 (Groningen), Mathematical Models in Ecology and Evolution Conference, poster
 Social Decision Making: Bridging Economics and Biology Workshop (Monte Verità), March 2011, poster
 ECCS 2010 (Lisbon), European Conference on Complex Systems, poster
 SUNBELT XXX (Riva del Garda), SUNBELT Conference of the International Network for Social Network Analysis, talk
 Evolution of Cooperation – Models and Theories Conference (Laxenburg), September 2009, poster
 CEC 2009 (Trondheim), IEEE Congress on Evolutionary Computation, talk
 ANTS 2008 (Brussels), 6th International Conference on Ant Colony Optimization and Swarm Intelligence, talk and poster
 ALIFE 2008 (Winchester), 11th International Conference on Artificial Life, talk
 GECCO 2008 (Atlanta), 10th Annual Conference on Genetic and Evolutionary Computation, talk
 AHS 2007 (Edinburgh), 2nd NASA/ESA Conference on Adaptive Hardware and Systems, talk
 AHS 2006 (Istanbul), 1st NASA/ESA Conference on Adaptive Hardware and Systems, talk

Teaching

“Mathematical Models of Social Evolution” (Summer 2025, 12 CM hours; Summer 2024, 12 CM hours; Summer 2022, 7 CM hours; Summer 2021, 14 CM hours), Toulouse Summer School in Quantitative Social Sciences
 “Population Demography and Evolutionary Ecology” (Fall 2021, 24 CM hours; Fall 2020, 24 CM hours; Fall 2019, 25 CM hours; Fall 2018, 18 TD hours), Toulouse School of Economics, Master 2 Environmental and Natural Resource Economics (ERNA), Ecology and Economics path (EEC). (Segment: Matrix population models)
 “Introduction to Ecology for Economists (M2EIGF2)” (Fall 2024, 15 CM hours; Fall 2023, 15 CM hours; Fall 2021, 15 CM hours; Fall 2020, 15 CM hours; Fall 2019, 15 CM hours; Fall 2018, 15 CM hours), Toulouse School of Economics, Master 2 Environmental and Natural Resource Economics (ERNA), Ecology and Economics path (EEC). (Segment: Introduction to population ecology)
 “Mathématiques TD 15h (L1EPDD1)” (Fall 2019, 15 TD hours), Toulouse School of Economics, undergraduate course (Licence 1)
 “Probabilités de gestion (LMAT04) - TD (L2GC062)” (Fall 2019, 7.5 TD hours, 3 groups), Toulouse School of Management, undergraduate course (Licence 2)
 “Probabilités TD (LEE2AED)” (Fall 2018, 15 TD hours, 2 groups), Toulouse School of Economics

“Advanced Behavioral and experimental Economics II (DUEB0309)” (Spring 2018, 6 CM hours), Toulouse School of Economics, Doctoral program DEEQA. (Segment: Social evolution theory)

“Economics and the environment for Economists (M2EIGA2)” (Fall 2017, 15 CM hours), Toulouse School of Economics, Master 2 Environmental and Natural Resource Economics (ERNA), Ecology and Economics path (EEC). (Segment: Introduction to population ecology)

“Economics and Evolution (DUEB0274)” (Spring 2017, 7.5 CM hours), Toulouse School of Economics, Doctoral program DEEQA. (Segment: Population and evolutionary dynamics)

“Evolutionary Dynamics: Game Theory (MA4454)” (Fall 2014, 8 hours), University of Lübeck, Mathematics in Medicine and the Life Sciences. (Invited lectures)

“Mathematics II (Graphs and networks)”, “Mathematics IV (Complexity)” (2006–2011), University of Lausanne, Faculty of Social and Political Sciences. (Teaching assistant for Henri Volken)

“Statistics I” (2006–2011), University of Lausanne, Faculty of Social and Political Sciences. (Teaching assistant for Jean-Philippe Antonietti)

“Electronic Circuits II Lab”, “Automatic Control Systems Lab”, “Artificial Intelligence” (2004–2005), Universidad Pontificia Bolivariana, Faculty of Electronics Engineering. Fixed-term part-time lecturer (Profesor de Cátedra)

Departmental and Professional Service

IAST management committee member (2018-2019, 2019-2020)

IAST recruitment committee member (2020-2021, 2021-2022)

IAST board member (since 2023-2024)

IAST/TSE Green Committee member (2019-2020)

Co-organizer of the Toulouse Summer School in Quantitative Social Sciences (2020-2021, 2021-2022, 2023-2024)

Journal referee for *The American Naturalist*, *Artificial Life*, *Behavioral Ecology and Sociobiology*, *Biology & Philosophy*, *BioSystems*, *Communications Biology*, *Communications Physics*, *Cognition*, *Current Opinion in Insect Science*, *Dynamic Games and Applications*, *Ecology Letters*, *Europphysics Letters*, *Evolution*, *Evolutionary Human Sciences*, *Games*, *IEEE Access*, *IEEE Technology and Society*, *IEEE Transactions on Automatic Control*, *IEEE Transactions on Evolutionary Computation*, *IEEE Transactions on Systems, Man and Cybernetics*, *Journal of the Association of Environmental and Resource Economists*, *Journal of Economic Behavior & Organization*, *Journal of Evolutionary Biology*, *Journal of Mathematical Biology*, *Journal of the Royal Society Interface*, *Journal of Theoretical Biology*, *Journal of Theoretical Politics*, *Management Science*, *Mathematical Biosciences*, *Mathematical Social Sciences*, *Nature Communications*, *Nature Computational Science*, *Neural Computing & Applications*, *Oikos*, *PeerJ*, *Philosophical Transactions of the Royal Society B*, *PLOS Computational Biology*, *PLOS ONE*, *Proceedings of the National Academy of Sciences*, *Proceedings of the Royal Society B*, *Royal Society Open Science*, *Science Advances*, *Scientific Reports*, *Swarm Intelligence*, *Theoretical Economics*, *Theoretical Population Biology*

Book proposal referee for *Oxford University Press*, *Princeton University Press*, *Springer Nature*

Grant proposal referee for *Research Foundation – Flanders (FWO)*, *European Research Council (ERC)*

Workshop co-organizer: IAST General Seminar (2021-2022); the 9th Toulouse Economics and Biology Workshop “Production and Sharing” (2022); the online seminar series of the [Evolutionary Models and Applications Group \(EMAG\)](#) (2021-2022), “The Biology and Economics of Mutualisms Workshop” at the Max Planck Institute for Evolutionary Biology (November 2017); Mini-symposium “Social Evolution in Subdivided Populations: Beyond the Usual Assumptions” at MME 2017

Member of the program committee at computer science conferences: ANTS (2016, 2012), ICANGA 2013, GECCO (2008–2013), CEC (2009–2011)

Member of the program committee at the 2022 International Conference on Social Dilemmas (ICSD) and the 20th Symposium of the International Society of Dynamic Games (2024)